Tolerance and Source-Normalization Robustness of the Native Weighted/Positive Budget Gap

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Abstract

A finite native-profile audit for the twin-prime dynamical bridge found a large source-budget separation between a signed weighted target and an all-positive control at one tolerance. We perform the next hostile test: the tolerance is varied over $\tau \in \{1/4, 1/2, 3/4\}$, both targets are evaluated in the same source prefix selected by the weighted target, and three source-side normalizations are compared. The finite budget

$$B_{k,\tau}(b) = \min\{c^T M_k c : \|V_k c - b\|_2 \le \tau \|b\|_2\}$$

has exact tolerance nesting, relative target homogeneity, threshold-prefix nesting, and common-prefix normalization invariance. On 18 frozen rows with 219 explicit shell targets, the common-prefix weighted/positive gap is above 10 in every row at every tolerance; its minimum is 155.1685, 69.9448, and 39.2637 at the three tolerances. The weighted budget is above $3 \cdot 10^{-5}$ under each of three normalizations in all 54 cases. This is a finite restricted robustness certificate, not an asymptotic prime estimate.

1 Question and finite model

The preceding releases measured literal source profiles for a finite prime-shell dynamical model. TPC-299 introduced the native source-budget frontier, and TPC-300 compiled selected points of that frontier as exact finite dual witnesses [1]. Those results used one normalized RMS tolerance and, in their main comparison, target-specific prefixes. A finite obstruction can be misleading if it is caused by a tolerance convention or by comparing two targets in different source spaces.

This paper isolates that vulnerability. Let S be a frozen shell, A the physical source-to-shell matrix, and U_k the first k literal cutoff profiles. Put

$$V_k = A^T U_k, \qquad M_k = U_k^T U_k.$$

For a nonzero target b and $0 < \tau < 1$, define

$$B_{k,\tau}(b) = \min\left\{c^T M_k c : \|V_k c - b\|_2 \le \tau \|b\|_2\right\}. \tag{1}$$

The source Gram is positive definite for every tested prefix. The targets are the TPC-299 signed minimum target and the all-one positive control; both have entries in $\{-1, 1\}$, so $\|b\|_2^2 = |S|$.

The source profiles are

$$\beta_z(t) = \lambda(t) - \sum_{\substack{d \le z \\ d \mid t}} \mu(d), \qquad z \in \{3, 5, 7, 11, 13, 17, 19, 23, 29, 31, 37, 41, 43, 47, 53, 59, 61\}. \tag{2}$$

All finite matrix entries are formed from rational arithmetic before the numerical frontier solve. The parent metadata contains a 1,380-edge grid count; the explicit shell lists used by this release contain 219 target edges. We retain both counts because they refer to different census fields.

2 Finite theorem layer

Proposition 1 (Tolerance nesting). *For fixed k and $0 < \tau_1 \le \tau_2 < 1$,*

$$B_{k,\tau_1}(b) \ge B_{k,\tau_2}(b)$$

whenever both frontiers are feasible.

Proof. The feasible set for τ_1 is contained in the feasible set for τ_2. Minimizing the same positive quadratic form over nested sets gives the inequality. □

Proposition 2 (Relative homogeneity). *For every nonzero scalar α,*

$$B_{k,\tau}(\alpha b) = \alpha^2 B_{k,\tau}(b).$$

Proof. The map $c \mapsto \alpha c$ bijects the feasible set for b with that for αb, since both the residual and its allowed radius are multiplied by $|\alpha|$. The objective is multiplied by α^2. □

Proposition 3 (Threshold prefix nesting). *Let*

$$r_k(b) = \min_c \frac{\|V_k c - b\|_2}{\|b\|_2}$$

and let $k_\tau(b)$ be the first prefix with $r_k(b) \le \tau$. Then $k_{\tau_2}(b) \le k_{\tau_1}(b)$ when $\tau_1 \le \tau_2$.

Proof. The column spaces are nested as k grows, so their distances from b are nonincreasing. Relaxing the tolerance can therefore only move the first feasible index to the left. □

Proposition 4 (Common-prefix normalization invariance). *Let $N_k > 0$ depend on the row and prefix but not on the target class. At a common prefix,*

$$\frac{B_{k,\tau}(b_w)/N_k}{B_{k,\tau}(b_+)/N_k} = \frac{B_{k,\tau}(b_w)}{B_{k,\tau}(b_+)}. \tag{3}$$

Proof. The positive scalar N_k cancels. The common-prefix condition is essential: if two targets are assigned different prefixes, their normalizers need not be the same. □

3 Hostile audit protocol

For each row and each $\tau \in \{1/4, 1/2, 3/4\}$, we compute all least-squares residual ratios along the 17-profile prefix ladder. Let $k_w(\tau)$ and $k_+(\tau)$ be the first feasible indices for the weighted and positive targets. The audit checks $k_+(\tau) \le k_w(\tau)$ and uses three contexts:

1. *common weighted prefix*: both targets use $k_w(\tau)$;
2. *target-specific prefixes*: each target uses its own first feasible prefix;
3. *full prefix*: both targets use all available profiles.

At every context we solve (1) by a monotone ridge-parameter bisection at 60-digit precision. The producer writes outward decimal intervals. The source normalizers are

$$N_\beta = \|\beta\|_2^2, \qquad N_{\text{mean}} = \frac{\operatorname{tr}(M_k)}{k}, \qquad N_{\text{first}} = M_k[1,1]. \tag{4}$$

The primary quantity is the weighted/positive ratio at the common weighted prefix. The target-specific and full-prefix results are controls, not substitutes for the primary comparison.

Table 1: Minimum weighted/positive budget gap over the 18 rows.

relative RMS τ	common prefix	full prefix	common > 10
1/4	155.1685273879	155.1685273879	18/18
1/2	69.9448236917	47.3501392957	18/18
3/4	39.2637006403	27.7392664825	18/18

4 Finite results

The project contains 18 rows, 219 explicit shell targets, three tolerances, three contexts, and two targets, for 324 frontier cases. Table 1 reports the minimum gap over rows. The full-prefix column is a secondary control.

The common weighted budget exceeds $3 \cdot 10^{-5}$ in 54/54 cases under each normalization in (4). The three common-prefix gap computations agree up to the replay precision in all 54 row–tolerance pairs, as predicted by Proposition 4. Full-prefix budgets are nonincreasing as the tolerance is relaxed in all 36 target–row checks, and the first-feasible-prefix ordering holds in all 54 row–tolerance checks.

The minima are not presented as asymptotic constants. They summarize a frozen finite atlas; in particular, the gap can vary by several orders of magnitude across rows. The useful conclusion is qualitative robustness: the separation does not disappear at the stricter or looser tested tolerance, nor when the two targets are placed in the same source space.

5 Reproducibility and claim firewall

The producer reconstructs the physical matrices and profile Grams from the frozen TPC-299 source-first engine. An independent checker, which does not import the TPC-301 producer, repeats every matrix build and frontier solve. An exact scalar stress suite verifies the theorem identities in ordinary and optimized Python modes. The canonical certificate records all case intervals and aggregate counts.

The finite theorem layer is proved exactly on the declared finite model. The atlas is numerically certified only relative to the declared rows, profiles, tolerances, target classes, and source normalizers. The current project does not prove a growing profile-budget bound, arithmetic L^2 estimate, fixed-power credit, full Gate B, or the twin-prime conjecture. The Session-named Route-A/Route-B evaluator files are absent from the checkout, so no official evaluator pass is asserted.

Conclusion

The TPC-300 finite native-budget obstruction survives the first natural hostile audit. A next meaningful step is to carry the same common-prefix and normalization protocol to growing shells, where the finite theorem identities remain valid but the budget and profile dimension must be controlled uniformly.

References

[1] Liang Wang. Native budget dual certificates, 2026. TPC-300 project release in the accompanying repository.